AzureMentor conversation

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Can you explain how vm works?

Yes - based on the context, a VM in Azure works as a machine attached to networking resources, and its network path can be either synthetic or accelerated.

1. A VM is created with networking attached to it. Azure creates a virtual network or lets you use an existing one, and a VM must have at least one NIC. A VM can have more than one NIC, and each NIC is connected to a virtual network and subnet [5].

2. By default, traffic uses a synthetic interface. When a VM is created, Azure creates a synthetic network interface for each virtual NIC. This interface is a VMbus device that uses the netvsc driver, and packets flow through the Azure host virtual switch and then onto the physical network [1].

3. If Accelerated Networking is enabled, a second interface is added. Azure creates an SR-IOV virtual function (VF) for each configured virtual NIC. In Linux, it appears as a PCI device and uses the Mellanox mlx4 or mlx5 driver [1].

4. Why this matters: most packets go directly between the Linux guest and the physical NIC, which lowers latency and uses less CPU than the synthetic path [1]. During Azure host servicing, traffic may switch back to the synthetic interface for about 30 seconds, sometimes a few minutes, so latency and CPU load can be higher then [2].

5. How to check which path is in use:
$ ethtool -i <interface name> | grep driver
If the driver is hv_netvsc, it's the synthetic interface. A VF driver name contains mlx, and ethtool -S eth<n> can show VF traffic counters [3].

If you want, I can also explain the difference between a VM, NIC, virtual network, and subnet using only this context.

Sources

[1] Virtual networks and virtual machines in Azure
https://learn.microsoft.com/azure/virtual-network/network-overview

[2] How Accelerated Networking works in Linux and FreeBSD VMs
https://learn.microsoft.com/azure/virtual-network/accelerated-networking-how-it-works

[3] How Accelerated Networking works in Linux and FreeBSD VMs
https://learn.microsoft.com/azure/virtual-network/accelerated-networking-how-it-works

[4] How Accelerated Networking works in Linux and FreeBSD VMs
https://learn.microsoft.com/azure/virtual-network/accelerated-networking-how-it-works